

Proctoring Using CrewAI

Project Scope & Implementation

A privacy-focused offline AI proctoring system that combines multimodal computer vision, audio analytics, ChromaDB-based RAG, and CrewAI multi-agent orchestration for incident analysis, reporting, and human review.

1. Project Scope

The scope of Proctoring Using CrewAI is to develop an intelligent examination monitoring platform capable of analyzing webcam and microphone data, identifying potentially suspicious activities, retrieving relevant examination rules, and using CrewAI agents to evaluate incidents and support human proctors.

- **Visual Monitoring:** Face tracking, gaze estimation, pose analysis, object detection, and lip-movement analysis.
- **Audio Monitoring:** Voice activity detection, speech transcription, and audio classification.
- **Incident Detection:** Combine visual and audio signals to generate structured incidents.
- **CrewAI Orchestration:** Coordinate specialized AI agents for incident analysis and reporting.
- **RAG with ChromaDB:** Retrieve relevant exam rules and historical incident context locally.
- **Human Review Assistant:** Present evidence and AI recommendations to a human reviewer for final assessment.
- **Offline Processing:** Keep core processing, embeddings, and data storage local for privacy.

2. Implementation

2.1 Frontend

Streamlit provides the monitoring dashboard with exam management, live proctor monitoring, webcam information, incident alerts, and incident reports.

2.2 Backend

FastAPI and Pydantic provide REST APIs, telemetry handling, incident processing, validation, and report generation.

2.3 Computer Vision

OpenCV, MediaPipe, and YOLOv8 are used for face tracking, gaze estimation, object detection, pose analysis, and lip-movement detection.

2.4 Audio Analytics

Whisper and Silero VAD are used for speech transcription and voice activity detection, supporting analysis of possible secondary voices or whispering.

2.5 ChromaDB and RAG

ChromaDB stores examination rules, transcripts, and incident context as local vector data. SentenceTransformers generates embeddings so relevant context can be retrieved during incident evaluation.

2.6 CrewAI Multi-Agent System

CrewAI coordinates role-based agents, tasks, and custom tools. A practical workflow can include Vision Analysis, Audio Analysis, Incident Evaluation, RAG/Policy Retrieval, and Human Review Assistant agents.

2.7 Human Review Assistant

The system presents the incident type, timestamp, evidence, retrieved rules, AI reasoning, and severity/recommendation to a human reviewer, who makes the final assessment.

3. End-to-End Workflow

```
graph TD
    A[Student Examination] --> B[Webcam + Microphone]
    B --> C[Vision & Audio Analysis]
    C --> D[Suspicious Event Detection]
    D --> E[Incident Generation]
    E --> F[ChromaDB RAG Retrieval]
    F --> G[CrewAI Multi-Agent Analysis]
    G --> H[Incident Explanation & Severity]
    H --> I[Human Review Assistant]
    I --> J[Final Incident Report]
```

4. Recommended GitHub Structure

```
Proctoring_Using_CrewAI/  
■■■ frontend/  
■ ■■■ app.py  
■ ■■■ pages/  
■ ■ ■■■ 01_Exam_Dashboard.py  
■ ■ ■■■ 02_Assessment_Session.py  
■ ■ ■■■ 03_Live_Proctor_Monitor.py  
■ ■ ■■■ 04_Incident_Reports.py  
■ ■■■ components/  
■■■ backend/  
■ ■■■ main.py  
■ ■■■ routes/  
■ ■■■ schemas/  
■ ■■■ vectordb/  
■■■ vision/  
■■■ audio/  
■■■ agents/  
■ ■■■ crew.py  
■ ■■■ agents.py  
■ ■■■ tasks.py  
■ ■■■ custom_tools.py  
■ ■■■ config/  
■■■ models/  
■■■ database/  
■ ■■■ chroma_db/  
■■■ requirements.txt  
■■■ .env.example  
■■■ .gitignore  
■■■ README.md
```

5. Technology Stack

- **AI Orchestration:** CrewAI
- **Frontend:** Streamlit
- **Backend:** FastAPI, Pydantic
- **Computer Vision:** OpenCV, MediaPipe, YOLOv8
- **Audio:** Whisper, Silero VAD
- **RAG / Vector Database:** ChromaDB, SentenceTransformers
- **Storage:** SQLite / PostgreSQL
- **Local ML:** PyTorch and local model weights

6. Short Project Description

Proctoring Using CrewAI is an intelligent examination monitoring system that combines computer vision, audio analytics, ChromaDB-based RAG, and CrewAI multi-agent orchestration. The system detects potentially suspicious activities, retrieves relevant exam rules and context, evaluates incidents through specialized agents, and provides evidence-based recommendations to a human reviewer through a monitoring dashboard.